And now, finally, looking at a dendrogram you can see why this is called hierarchical clustering: it keeps track of a hierarchy of clusters going (top-to-bottom) from a single cluster containing the whole dataset, to `N` singletons, proto-clusters with a single dataset point in them. Moving from the top of the dendrogram to its bottom means virtually exploring all the possible values for ε, and the flat clusterings associated with those values: when, in figures 12.17, 12.19 etc., we chose `ε=0.5` or `ε=0.6`, we were figuratively cutting a section of the dendrogram and taking a peak at the clusters formed at that level. As we have seen, though, cutting such a section, with a line perpendicular to the ε axis (meaning, a line where ε is constant through the whole dataset), doesn't work for our non-uniform dataset.

The great thing about having a hierarchy of clusters, though, is that we don't have to cut this dendrogram at the same height for the whole dataset! In other words, we can use different values of ε in different branches of the dendrogram. How do we decide which branches and what values? Well, with 2D datasets we can be guided by our intuition, but in higher dimensions, it can stem from domain knowledge, or be derived by some metric: for instance, in our example, we can consider the partitioning after the first split in the dendrogram, `CE` and `CF`, and compare their average density: since density is clearly very different between the two branches, we can come up with two different values for ε in each of them, based on the statistics of each subset: higher where density is lower (or, equivalently, where the average reachability distance is higher).